WANKOE Line Design Report

Limestone processing line — complete design, calculation and verification record

NOEZYS · AI Innovation Lab — full report, chapters 0–10 + annexes, 2026-08-14 — confidential

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Contents

0 Executive summary

The whole line on one page — what is sold, what is proven, what remains open.

The WANKOE line converts a single quarry feed into four sold products. At the configuration adopted on 2026-08-14 — the redesigned zone 1.3 operating in two modes behind a grits diverter, dedicated AgLime campaigns in zone 1.2, and the impactor at its low-speed setting — the annual plan serves **every market exactly**:

Product	Sold (t/y)	Status	Zone utilization
KFS 20/35 (kiln, firm)	85 000	envelope compliant	zone 1.1 — 71.2 %
FeedLime grits 2/4 (firm)	40 000	D6 compliant	zone 1.3 — 60.2 % (2 466 h mode G + 1 144 h mode F)
FeedLime fines 0/1.5 (objective)	60 000	96.6 % < 1.7 mm	
AgLime 0/1.7	135 000	98.4 % < 1.7 mm	zone 1.2 — 40.5 %

The single open cost is the disposal of surplus crushed fines: **13 816 t/y of 0/20 still go to landfill** at today's measured feed. The governing indicator is the *KFS Yield*, defined by the client and computed on every engine run:

$$Y_{KFS} = \frac{\dot{m}^{wet}_{KFS}}{\dot{m}^{wet}_{pivot}} \qquad Y^{req}_{KFS} = \frac{T_{KFS}}{T_{KFS} + D_{0/20}} \tag{0.1}$$

Today the line realizes $Y_{KFS} = 24.88\,\%$ against a zero-landfill requirement of $25.93\,\%$. Closing the 1.05-point gap is the object of the adopted quarry works specification (chapter 8): at the target inlet curve the balance closes exactly — zero landfill with a 20 % AgLime-market buffer held open as the shock absorber.

> **Confidence statement.** The zone-1.1 calculation chain passed a six-action internal verification program: blind independent recalculation (full concordance, 25 scalars + 29-point curve), literature anchors (7/7, sources cited in-test), 580-scenario invariant sweep, a Monte-Carlo uncertainty band (Y_{KFS} = 24.3 % [P10 22.4 – P90 26.0]), numerical-grid convergence (adopted ×2 grid), and a two-lens adversarial formula audit (zero sign or unit errors; three boundary-condition fixes). Meta-score **94 %** — the internal ceiling; the remaining points require the vendor gradation test, drop-weight and sieve tests, repeat belt-cuts, and the design-curve reconciliation with the engineering contractor.

The redesign in one paragraph

The as-built zone 1.3 ground 2.83 tonnes of fines for every tonne of grits and failed the adopted grits specification (15.4 % below 2 mm against a 15 % limit). The adopted C1 circuit — two-stage smooth rolls with immediate product relief on a double-deck screen pair — cuts the ratio to 0.84, passes the specification, and, operated in two modes behind the grits diverter DV.GF, decouples the grits and fines order books entirely. Purchase specifications: RC.1 32 t/h; RC.2 two units of 22 t/h each, gap range down to 1.5 mm (mode F), all [H] coefficients pending the vendor gradation test.

1 Conventions and definitions

The rules every number in this report obeys — flow bases, stream names, the yield indicator, and the computation grid.

One principle frames everything: **the result of this project is the mathematical model, never a document** — this report is a printout of model output, dated and replayable, and the model outlives it. The engine computes one deterministic "photo" per scenario, never optimizes on its own and imposes no operating choice; sweeps and plans are explicit, user-configured calls. The line is a new process under construction, and the model's ultimate purpose is to **confirm the line design and the machine selection**. The only plant data that will ever exist are belt-cut PSD analyses at the primary crusher outlet; downstream behavior rests on the validated engineering models of chapter 3 and their documented hypotheses — a constraint the verification program of chapter 9 was built around.

1.1 The total-flow rule (wet basis)

Every line feed rate in this report — and in the model's data file — is the **total flow on the wet basis**: dry solids plus their water, exactly as a belt weigher would report it. The rule was ratified by the client on 2026-08-14 and is load-bearing: zone 1.1 is fed at 250 t/h wet, the zone-1.2 reclaim runs at 100 t/h wet, and the zone-1.3 FeedLime feed rates (32.1 t/h in mode G, 25.05 t/h in mode F) are wet as well. The engine converts to dry solids at each zone boundary, because dry solids are the conserved quantity (water leaves only at the dryer DY.03):

$$q_{dry} = q_{wet}\left(1 - \frac{w}{100}\right) \tag{1.1}$$

with w the wet-basis moisture in percent (7 % as measured with the 2026-08-08 belt cut; 0.5 % downstream of the dryer). Presented rates are quoted in total (wet) flow unless they are explicitly labeled dry; in the calculation chapters the wet figure comes first and the dry figure follows in parentheses wherever a balance is being closed.

1.2 Stream naming convention

The client ratified a strict naming rule on 2026-08-14, after one table had blurred it: **zone 1.3 produces DRY products only** — FeedLime grits 2/4, FeedLime fines 0/1.5 and UltraFin, all at 0.5 % moisture. **AgLime 0/1.7 is a WET zone-1.2 product**; zone 1.3 never produces AgLime. The zero-waste rule that sends the FeedLime-fines surplus into the AgLime sales channel is a **commercial routing at loadout** — a line in the planning's sales view, never a production stream. Tables in this report therefore say "redirect eligibility (≥ 95 % < 1.7 mm)" when they qualify a zone-1.3 fines stream against that routing, and reserve the words "AgLime spec" for the zone-1.2 product itself.

Machine tags follow the specification codification (CR.5009, SR.5007, CR.5011, SR.5105, SR.5111, CR.5113, SR.5115, DY.03, SP.36, CL.38); the C1 redesign units carry the working tags RC.1, RC.2, SC.A, SC.B and DV.GF until the final codification (an SN.2x/CR.2x proposal to the engineering contractor is a chapter-10 item). The jaw crusher CR.5003 sits upstream of the pivot measurement point and is deliberately outside the flowsheet — its parameters feed the reference-curve calibration script only.

The line map, for orientation:

Zone	Function	Machines	Products (state)
1.1	crushing / screening block ("zone 1")	CR.5009 → SR.5007 ⇄ CR.5011	KFS 20/35 (wet); 0/20 stream to stockpile (wet)
1.2	reclaim and wet finishing	SR.5105; SR.5111 → CR.5113 ⇄ SR.5115	AgLime 0/1.7 (wet); FeedLime 6/20 interstage (wet)
1.3	drying and dry finishing (C1)	DY.03 → RC.1/RC.2 ⇄ SC.A/SC.B (+ SP.36/CL.38)	FeedLime grits 2/4, FeedLime fines 0/1.5, UltraFin (all dry)

1.3 Operating modes

Each zone runs one of a small set of explicit operating modes; the engine fails fast on an unknown mode and never chooses one by itself:

Zone	Mode	Meaning	Status at the adopted plan
1.1	1A	production: KFS 20/35 to stock + 0/20 to stockpile	the plan's only zone-1.1 mode
	1B	full-crushing stock campaign (all flow through the impactor)	unusable at 250 t/h with CSS 30 — and no longer needed
1.2	2A	reclaim split at 6 mm: FeedLime 6/20 + AgLime loop	demand-driven hours (strict c2 rule)
	2B	rain mode: FeedLime = whole reclaim (mass identity)	0 h required at the adopted plan
	2C	dedicated AgLime campaign: everything reclaimed leaves as AgLime	completes the AgLime market (chapter 5)
1.3	G	grits production: 2/3.75 band to the grits product chain	2 466 effective h/y
	F	fines campaign: 2/3.75 diverted into the RC.2 regrind circuit, RC.2 at the 1.5 mm mode-F gap	1 144 effective h/y

1.4 Size curves and their bookkeeping

Every particle-size distribution is a **cumulative % passing** curve on a logarithmic mesh axis, stored as a value table with a declared interpolation mode (log-linear throughout this project) — the specification's golden rule 2, which forbids implicit interpolation of vendor or measured curves. The conserved quantity of every balance is **dry solids**; water is carried alongside each stream as wet-basis moisture and leaves the line only as vapor at the dryer DY.03. Both the dry-solids and the water balance of every zone are asserted on every engine run at a strict tolerance (the realized closure gaps of the runs in this report are below 2×10^{-6} relative).

1.5 The KFS Yield indicator

The project's governing indicator was introduced by the client on 2026-08-14 and its definition was arbitrated in four explicit sub-decisions: (1) the numerator is the **whole KFS product stream** (conveyor BC.5013 to stock), always reported together with its real 20/35 PSD (in-cut / below / above), never a hypothetical in-spec fraction; (2) the basis is **wet over wet** — the commercial basis, what belt weighers see; (3) the denominator is the **pivot feed**, measured downstream of the primary crusher CR.5003, the project's one standard measurement point; (4) the usage is an indicator with a **dynamic target**: the required-for-zero-landfill yield $Y_{KFS}^{req} = T_{KFS}/(T_{KFS} + D_{0/20})$, where $D_{0/20}$ is the annual downstream demand for crushed 0/20, with an engine alert whenever the realized value falls below the required one. Both values appear in every scenario result and in the annual plan.

1.6 Computation grid and presentation grid

All particle-size computations run on the **×2 refined grid** adopted by the client on 2026-08-14 (engine key `computation_grid_refinement = 2`): one geometric midpoint is inserted in every interval of the 29-sieve specification series, giving 57 computation meshes, converged to better than 0.05 point of KFS Yield against a ×4 grid (chapter 9, action 5). The specification sieve series remains the **presentation format** — every curve printed in this report is read off at the spec sieves. Chapter-9-era reference tests pin refinement 1 so the model still reproduces the specification's own numerics on demand.

1.7 Symbols, units and flags

Units follow the project data file throughout: t/h, mm (µm where stated), kW, % by mass, kWh/t, °C, h; moisture is wet-basis. Coefficients carry their provenance flag: **[ref.]** traceable to literature or vendor documents, **[H]** a documented hypothesis awaiting a measurement, **[F]** fitted on the specification's reference case. Every [H] coefficient that a result depends on is flagged where the result is presented.

2 Design basis and decision trail

What the line must sell, what the redesign must satisfy, and how every arbitration is recorded.

2.1 Production targets and market caps

The commercial frame, as encoded in `production_targets` (all values client-arbitrated, final states dated 2026-08-13/14):

Product	Target (t/y)	Nature	Note
KFS 20/35	85 000	FIRM	kiln contract — absolute priority
FeedLime grits 2/4	40 000	FIRM	80 000 t/y documented as the extension case (D1 sizing case)
FeedLime fines 0/1.5	60 000	objective + market cap	production objective since 2026-08-14 (was a passive cap)
AgLime 0/1.7	135 000	market cap	loop co-production + dedicated 2C campaigns + fines redirect
UltraFin < 100 µm	4 000	capped by natural fines content	optional extraction (D7)

Two commercial rules close the balance: the **zero-waste redirect** (2026-08-13) — any fines surplus beyond the 60 kt market is redirected into the AgLime sales channel at loadout, the AgLime loop producing only the complement to the cap — and the **landfill ruling** (2026-08-13, final): no market exists for crude 0/20, so any 0/20 produced beyond the downstream reclaim goes to landfill and is reported as a net financial loss with an alert, never as a sale.

The product specifications, as encoded in `output_products` and evaluated on every run:

Product	Cut	State	Acceptance criterion
KFS	20 / 35 mm	wet	30/55/15 envelope: ≤ 30 % below cut, ≥ 55 % in cut, ≤ 15 % above cut (validated 2026-08-08; supersedes the unworkable 15 % total out-of-cut reading — Q4)
AgLime	0 / 1.7 mm	wet	≥ 95 % passing 1.7 mm (client 2026-08-12 — the first numeric AgLime spec)
FeedLime grits	2 / 4 mm	dry (0.5 %)	D6 envelope: ≤ 15 % below 2 mm, ≤ 5 % above 4 mm (client 2026-08-14)
FeedLime fines	0 / 1.5 mm	dry (0.5 %)	no numeric spec; redirect eligibility ≥ 95 % < 1.7 mm when routed to the AgLime channel
UltraFin	0 / 0.1 mm	dry	no numeric spec; NOT CERTIFIED until Φ measured
Sliver 1.5/2	1.5 / 2 mm	dry	present only when the sliver diverter is on "extract" — normally absent (regrind)

2.2 Zone-1.3 design basis D1–D8

The client judged the as-built zone 1.3 badly dimensioned and ordered a from-scratch redesign on 2026-08-14. The design basis was frozen element by element, each row validated individually:

#	Item	Requirement
D1	Duty	one machine park for BOTH 40 000 and 80 000 t/y grits; sizing case 80 kt — only rates, settings and hours differ
D2	Fixed equipment	dryer DY.03 acquired: 30 t/h at outlet (32.1 t/h wet feed), the single drying line; everything else free
D3	Fines constraint	fines/grits ≤ 1.25 FIRM (80 kt through the single dryer); ≤ 1.0 objective; grits share ≥ 44 % of dried product
D4	Design feed	sized on the measured chain's FeedLime 6/20; every candidate verified on the quarry-target variant too
D5	Shift frame	3 × 8 h continuous, 7 500 h/y ceiling, 80 % availability = 6 000 effective h/y
D6	Grits quality	2–4 mm: > 4 mm ≤ 5 %; < 2 mm ≤ 15 %; moisture 0.5 % — adopted 2026-08-14, both thresholds adjustable parameters
D7	Co-products	fines 0/1.5 sellable (60 kt market + redirect-compatible); UltraFin optional
D8	Commercial frame	zero unsellable product; 0/20 landfill to avoid by construction; candidates judged on the FULL line balance

2.3 Document authority: specification, PFDs, expert book

Three layers of source documents govern the model, in a client-ruled order of authority. The **specification** (nine chapters plus preamble, v. 2026-08-08) defines the models M1–M8, the flowsheet, the products and the reference cases; its conformity is tracked requirement by requirement in the conformity matrix. The three **NACO/Carmeuse PFDs** received on 2026-08-12 (zone 1.1 REV15, zone 1.2 REV18, zone 1.3 DBR) **prevail over the specification wherever they diverge** — the client's authority ruling of the same day — with one exception: in zone 1.3 the PFD's DV.10 and ML.30 "Unirotor" were deleted by the client, and since 2026-08-14 the whole zone is superseded by the C1 redesign of chapter 6. The ruling forced the complete rebuild of zone 1.2 to the REV18 topology (single-deck 6 mm FeedLime cut; open + closed 1.7 mm AgLime loop) presented in chapter 5. The **expert book** (23 pages, 22 references, received 2026-08-11) derives every machine model from first principles with a provenance status per coefficient; it confirmed the implemented architecture and closed three of the open arbitration questions (Q2, Q6, Q8). Four of its issues await an expert clarification note (imperfection convention and value; the book's CR.5009 worked-example F80 120 mm vs the measured 180.6; S_att 0.206 vs the fitted 0.171; CR.5011 capacity) — all tracked in the register.

2.4 The 12-question arbitration round

The client answers best to one question at a time with proposed options; the standing round of twelve structural questions (opened 2026-08-10) reads as follows today:

Q	Question	State

Q1	screen imperfection value	CLOSED 2026-08-10 — I = 0.15 from literature [H], sieve test to refit
Q2	Bond work index 12.54 vs 13.8 (short-ton reading)	CLOSED 2026-08-11 — 12.54 kWh/t confirmed [ref.] (expert book)
Q3	imperfection-remap ratification + convention nuance	OPEN — tied to the expert clarification note and the sieve test
Q4	KFS tolerance vs envelope (the nulled 15 % out-of-cut)	OPEN
Q5	spec §9.3 fines figure 19.9 t/h (cannot close mass)	OPEN — model enforces closure
Q6	SP.36 cut size	CLOSED 2026-08-11 — 65 µm (0.65·d97, Stokes; expert book)
Q7	KFS 85 kt securing lever (zone-1.1 hours)	CLOSED 2026-08-13 — Saturday extension, ceiling 2 400 h
Q8	CR.5011 real capacity	CLOSED 2026-08-11 — HAZEMAG AP-S 1010 docs: 75–90 t/h, motor 132 kW
Q9	hosting	CLOSED 2026-08-09 — deployed with access control
Q10	repository location	OPEN (deployment-safety caution)
Q11	French web-UI exception	CLOSED 2026-08-09 — REJECTED; English everywhere
Q12	b_j for soft limestone	OPEN — on hold pending the commissioned hardness note; drop-weight test decides

2.5 The decision-register method

This is an evolving project, never a frozen one. Every change lands through the data file first and code second, and every arbitration receives a dated row in the decision register (`docs/spec-conformity-matrix.md`) stating **who decided**: the client, a delegated hypothesis, or an expert proposal pending ratification. The register is the project's memory and the authority whenever an older document diverges from it; this report quotes register rows with their dates wherever a figure is a decision rather than an engine output. The 2026-08-13 and 2026-08-14 rows carry the whole story of the adopted configuration; the most load-bearing of them are condensed below.

Four standing client rules govern how work lands. **Nothing hardcoded**: every parameter is adjustable in the data file without reprogramming — the redesign of chapter 6 was wired entirely through data plus one flowsheet variant. **Hours follow the targets, never the reverse**: annual production targets are the inputs, operating hours the computed outputs, shift regimes only capacity ceilings. The specification's **golden rules**: an undefined symbol becomes a question, never a guess; vendor curves are value tables with a declared interpolation mode; data is separated from code. And **maximum rigor and honesty**: expert panels and adversarial verification at every significant step, with findings disclosed even when uncomfortable — the fines-cascade episode of §6.4 and the grid-bias finding of §9.5 are this rule in action. A companion traceability rule (2026-08-11) requires every issued datasheet and dossier to carry an engine-run provenance footer, and any figure not produced by an engine execution to be flagged; the extraction scripts are archived in the repository so any engineer can replay the numbers without assistance.

2.6 The issued document record

The client-facing paper trail this report consolidates: technical dossier **DT-001** (zone 1 complete, eleven machine datasheets at the client-validated inputs, each datasheet validated in review before issuance, tracked in the dossier register — revision pending on the C1 adoption); the zone-1 **calculation-sheet document** issued for external design review (2026-08-12, thesis-grade, three machines plus zone balance and weekly translation); the zone-1.2 calculation document, withdrawn when the PFDs superseded its topology and pending re-issue on the REV18 rebuild; the **quarry works specification** (chapter 8, issued 2026-08-14); the zone-1.3 **design-basis, panel-diagnosis and candidate notes** plus the block diagram (as-built and C1) in the design dossier; and the six verification action reports of chapter 9. Every issued figure traces to an engine run through the provenance footers.

Date	Decision (condensed)	Decided by
2026-08-13	Q7 closed — Saturday extension: zone 1.1 ceiling 2 000 → 2 400 h/y; 85 kt KFS firm commitment feasible	the client
2026-08-13	FeedLime fines market defined: 60 000 t/y sellable maximum	the client
2026-08-13	Zero-waste rule: fines surplus redirected into the AgLime sales channel; loop makes only the complement	the client
2026-08-13	Dryer capacity defined: 30 t/h AT THE OUTLET = 32.1 t/h wet feed	the client
2026-08-13	Reference zone-1 settings g = 60 / CSS = 30 adopted (0/20-balance campaign)	the client
2026-08-13	0/20 excess = LANDFILL, a net financial loss — no crude market exists (supersedes the same-day crude-sales rule)	the client
2026-08-14	Zone-1.3 redesign launched; design basis D1–D8 frozen	the client
2026-08-14	D6 grits envelope encoded — the as-built circuit FAILS it (15.4 % < 2 mm)	the client (spec) + engine (finding)
2026-08-14	C1 validated as lead candidate, then FORMALLY ADOPTED as the reference configuration	the client
2026-08-14	Sliver 1.5/2 arbitration: REGRIND through RC.2 (two-position diverter, reversible)	the client
2026-08-14	Screen arrangement: 2+2 — SC.A recycle cuts 8/3.75, SC.B product cuts 2/1.5	the client
2026-08-14	RC.1 purchase specification raised to 32 t/h (closes the D4 quarry-feed finding)	the client
2026-08-14	AgLime dedicated 2C campaigns wired into the planning (landfill 144.2 → 58.4 kt/y at that epoch)	the client (rule) + engine
2026-08-14	Impactor v = 35 → 30 m/s — the last settings headroom against landfill	the client
2026-08-14	KFS Yield indicator defined (4 arbitrated sub-decisions, §1.3)	the client

2026-08-14	×2 computation grid adopted; converged figures rebased (KFS Yield 24.59 → 24.88 %)	the client
2026-08-14	Fines 60 kt becomes a production OBJECTIVE; grits diverter DV.GF; two-mode zone 1.3; second RC.2 unit; mode-F gap 1.5 mm	the client + engine (feasibility)
2026-08-14	Total-flow rule ratified: all feed rates wet (dry + water), as belt-weighed	the client
2026-08-14	Quarry works target adopted: 40.1 % passing 20 mm at the zone-1 inlet, 20 % AgLime flex margin	the client

3 The calculation models M1–M8

Eight machine models, their exact formulas, their provenance, and the seven literature anchors that pin them.

All machine physics lives in eight pure functions (one model, one function, coefficients in the data file only). The scientific review (`docs/model-science-review.md`) grades each on provenance: **A** — established law in its exact textbook form; **B** — established family with a spec-specific parameterization; **C** — documented, bounded, fitted phenomenological hypothesis.

A scenario execution composes them in a fixed pipeline: the feed builder assembles the pivot PSD from the measurement plus the H-FEED completions and refines it to the computation grid; each zone function then chains its machines — M1/M2 at every crusher, M3/M4 at every screen deck, M5 feeding M1's slope at the impactors, M6 at the dryer, M7 at the rolls, M8 at the classifier — iterating the closed loops to their fixed points; and the scenario layer closes the dry-solids and water balances, evaluates every product against its specification, computes the indicators and emits the alerts. Nothing else happens: no hidden state, no optimization, one photo per call.

3.1 M1 — Crusher product: truncated Rosin–Rammler (grade A)

$$P(x) = 1 - \exp\left[-\left(\frac{x}{x_c}\right)^n\right], \qquad x_c = \frac{x_{80}}{(\ln 5)^{1/n}} \tag{3.1}$$

Rosin & Rammler (1933), the universal representation of comminution products; the $\ln 5$ anchor follows exactly from requiring $P(x_{80}) = 0.8$. The product is truncated at $1.7\,x_{80}$ with mass rescaling (spec parameter `trunc_factor`), and feed already finer than the product size bypasses unbroken — the standard classification device of Whiten-type crusher models. Disclosed nuance (adversarial audit, 2026-08-14): after truncation the delivered value at the nominal x_{80} is ≈ 0.83, stated in the data notes. The slope n is machine-specific ([H] for the smooth rolls until the vendor gradation test).

3.2 M2 — Comminution power: Bond's third theory (grade A)

$$W = 10\,W_i\left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}}\right)\ \mathrm{kWh/t} \quad (\mu\mathrm{m}), \qquad P_{inst} = \frac{W\,Q}{\eta_m} \tag{3.2}$$

Bond (1952), the industry standard for seventy years, implemented in its exact textbook form and hand-checked in the tests. $W_i = 12.54$ kWh/t is **[ref.]** since Q2 closed on 2026-08-11 (the expert book's referenced Belgian-limestone value; consistent with the measured limestone range 12.6–13.9 kWh/t); $\eta_m = 0.75$. Accepted accuracy for impact crushers is ±25–30 % — a limit of the law, adequate for the design-margin use this project makes of it.

3.3 M3 — Screen partition: efficiency curve (grade B)

$$\rho(x) = \frac{1}{1 + (d_{50c}/x)^s}, \qquad d_{50c} = a\, k_d, \qquad s = \frac{\ln 9}{\ln(1/(1-I))} \tag{3.3}$$

The partition-curve standard of mineral-processing simulation (Reid/Plitt logistic family, Lynch-school sharpness through an imperfection I; the spec labels it "Karra"). The formula as written in the spec contradicted its own narrative and was arbitrated on 2026-08-08 (option A: the narrative wins); $I = 0.15$ was adopted from literature on 2026-08-10 **[H]**, pending a screen-product sieve test. **Disclosed convention nuance** (action 2, 2026-08-14): with the spec's sharpness formula, inputting $I = 0.15$ yields a partition curve whose *classic* imperfection $(d_{75} - d_{25})/(2\, d_{50})$ is $\sinh(\ln(1/(1-I))/2) \approx 0.081$ — the implemented screens are about twice as sharp as a classic $I = 0.15$ screen, close to Karra 1979's fixed-sharpness family. The convention is client-arbitrated and stated wherever it matters; it belongs to the Q3 sieve-test closure.

3.4 M4 — Screen area: VSMA sizing (grade B)

$$A = \frac{U\, f_p}{Q_b\, f_0}, \qquad Q_b = 14\, a^{0.6} \tag{3.4}$$

The VSMA handbook method, the industry's screen-sizing standard; $14\, a^{0.6}$ is the spec's fit of the basic-capacity curve and the global correction $f_0 = 0.347$ **[H]** (fitted on SR.5007) absorbs the usual factor stack, with peak factor $f_p = 1.2$ **[H]**. Sizing-level accuracy ±20–30 % — exactly the design-confirmation use. Under rain the basic capacity derates by the factor 0.75 **[H]**.

3.5 M5 — Impact breakage: JKMRC t10 (grade A)

$$E_{cs} = \frac{v^2}{7200}, \qquad t_{10} = A_j \left(1 - e^{-b_j E_{cs}}\right), \qquad n = \left(\frac{30}{t_{10}}\right)^{0.30} \tag{3.5}$$

The JKMRC breakage model (Napier-Munn et al.), backbone of JKSimMet. $v^2/7200$ is exact physics — the rotor-tip kinetic energy per unit mass converted to kWh/t (proven an identity in action 2). $A_j = 60$ and $b_j = 0.8$ are literature defaults **[H]**; the JKTech database trend puts soft limestone at $A{\cdot}b \sim 60$–80, so the current $A{\cdot}b = 48$ is a conservative hard-side choice — the drop-weight question Q12 remains open. The $t_{10} \to n$ mapping is the spec's bounded, monotone simplification.

3.6 M6 — Drying: mass and energy balance (grade A)

$$Q_{th} = \frac{\dot{m}_w\, (L_v + c_e\, \Delta T_e) + \dot{m}_s\, c_s\, \Delta T_s}{\eta_{th}}, \qquad V_{drum} = \frac{\dot{m}_w}{I_{ev}} \tag{3.6}$$

First-principles thermodynamics: latent heat 2 257 kJ/kg, sensible heats, thermal efficiency $\eta_{th} = 0.6$ **[H]**, evaporation intensity 45 kg/m³·h **[H]** for drum sizing (Perry's rule of thumb). Water closure is asserted on every run. This is the most certain model of the chain (it reproduces the spec's reference figures to three digits); the adversarial audit added the boundary fix that a no-drying pass reports exactly zero duty.

3.7 M7 — Bed roller mill / smooth rolls (grade C — the honest weak point)

A two-component phenomenological model: material coarser than the gap is compressed to a Rosin–Rammler product at $x_{80} = g$ (slope n_{comp}, per-pass reduction capped by λ_{comp}, hypothesis H-M7-1), and a fraction S_{att} of each pass reports to attrition fines with their own RR parameters (hypothesis H-M7-2); sub-gap material bypasses unbroken — pinned *at the gap* even in the capped regime since the 2026-08-14 audit fix. The structure is validated by literature for toothed double rolls (Thiere, TU Freiberg 2020), which raised the grade from C to B− for ML.26; for the C1 smooth rolls the coefficients ($n_{comp} = 1.8$, $\lambda_{comp} = 2.2$, $S_{att} = 0.06$) are **[H]** and are trial trigger #1: the vendor gradation test on WANKOE 6/20 fixes them before any purchase. A vendor product curve, when available, plugs directly into the data (`product_curve_table` slot).

3.8 M8 — Air classification and cyclone (grade A/B)

$$\dot{m}_{fine} = \dot{m}_{feed}\, \Phi(< cut)\, \eta_{cl}, \qquad d_{50} = \sqrt{\frac{9\, \mu\, b}{2\pi N_e\, v_{in}\, (\rho_p - \rho_a)}} \tag{3.7}$$

The standard efficiency-recovery classifier form, mass-exact per size interval, and the Lapple (1951) cyclone model in its exact form ($d_{50} = 4.2$ µm at the adopted Stairmand geometry, $b = 0.14$ m, vendor drawing to confirm). Φ at the 65 µm cut (Q6 closed 2026-08-11) comes from the modelled curve until measured and every UltraFin figure is flagged NOT CERTIFIED accordingly; $\eta_{cl} = 0.75$ **[H]**. The sub-4 µm cyclone tail requires a polishing bag filter that is absent from the flowsheet — a standing design-review item.

3.9 Literature anchors — 7/7 pass

Action 2 of the verification program (chapter 9) confronted each model with its source literature, every expected value hand-computed into the permanent test module `tests/test_literature_anchors.py` with the source cited:

Model	Anchor (source)	Result
M2 Bond	worked case W_i 12.74, 25 → 3 mm → W = 1.520246 kWh/t (Wills)	exact to 1e-5
M5	$E_{cs} = v^2/7200$ exact physics identity; v = 30 → 0.125	exact to 1e-12
M5	JKMRC t_{10} = 5.709755 % (Napier-Munn 1996)	exact to 1e-5
M3	logistic partition quartiles x_{25}/x_{75} analytic (Reid/Plitt; King)	both reproduced
M3	sharpness s = ln 9 / ln(1/(1−I)) = 13.5198 at I = 0.15	exact
M4	VSMA effective capacity vs published Factor-A: 29.3 vs 30.8 (20 mm); 41.0 vs 39.5 (35 mm)	within 5 %
M1	truncated RR $P_t(x_{80})$ = 0.830814 (hand exponentials) + hard truncation at 1.7· x_{80}	exact to 1e-5

One disclosed incident does the program credit: the checker's first 4-digit hand value for the M1 anchor (0.830783) disagreed with the engine; the high-precision recomputation confirmed **the engine** (0.830814) — the independent check corrected the checker, recorded verbatim in the test docstring.

3.10 Model verdict and the upgrade path

The scientific review's one-line verdict per model, with the single measurement or vendor input that would upgrade it:

Model	Grade	The one thing that would upgrade it
M1 Rosin–Rammler	A	n per machine from vendor gradation curves
M2 Bond	A	accept ±25–30 % — intrinsic to the law (site Bond test optional)
M3 partition	B	screen imperfection from a product sieve test (Q3)
M4 VSMA	B	installed areas from design drawings
M5 JKMRC	A	A_j / b_j from a drop-weight test (Q12)
M6 drying	A	—
M7 bed mill / smooth rolls	**C**	vendor performance curve (product_curve_table slot ready)
M8 classifier / cyclone	A/B	Φ measurement at the cut; cyclone inlet width from the vendor drawing

3.11 Key coefficients and their provenance

The load-bearing coefficients of the chain, from the calibration block of the data file (every one adjustable without reprogramming — the client's rule 1):

Symbol	Meaning	Value	Status	Used by
Wi	Bond work index	12.54 kWh/t	[ref.] — Q2 closed	M2
η_m	mechanical efficiency	0.75	project	M2
A_j / b_j	JKMRC breakage parameters	60 % / 0.8	[H] — conservative hard-side; Q12 open	M5
I_dry / I_rain	screen imperfection	0.15 / 0.9	[H] — Q1 arbitrated; sieve test to refit	M3
k_d	screen d50c factor	1.0	[H]	M3
f0 / f_p	VSMA global correction / peak factor	0.347 / 1.2	[H] — f0 fitted on SR.5007	M4
L_v, c_e	latent heat, water heat capacity	2 257 kJ/kg, 4.18	physics	M6
η_th / I_ev	dryer thermal efficiency / evaporation intensity	0.6 / 45 kg/m³·h	[H]	M6
λ_comp, n_comp, S_att (ML.26)	as-built mill hypotheses H-M7-1/2	4.11 / 0.86 / 0.171	[H][F] — fitted on the spec's reference case	M7

λ_comp, n_comp, S_att (RC.1/RC.2)	smooth-roll hypotheses	2.2 / 1.8 / 0.06	[H] — vendor gradation test = trial trigger #1	M7
η_cl	classifier extraction efficiency	0.75	[H] — flagged optimistic	M8
N_e, b, ρ_p	cyclone turns, inlet width, particle density	5 / 0.14 m / 2 700 kg/m³	[H] — Stairmand assumption, vendor to confirm	M8
Φ(<cut)	natural fines content below the 65 µm cut	null	TO BE MEASURED — UltraFin not certified until then	M8

4 Zone 1.1 — complete calculation

From the 250 t/h wet pivot feed to KFS 20/35 and the 0/20 stream, machine by machine, at the adopted settings g = 60 / CSS = 30 / v = 30.

All figures in this chapter are one engine execution of `run_scenario` at the default parameter set (measured 2026-08-08 feed curve, mode 1A, dry weather, ×2 grid). Zone 1.1 is the crushing/screening block — what the client means by "zone 1": the pivot feed enters the roll crusher CR.5009, its product joins the recirculating load on the double-deck screen SR.5007, the top-deck oversize closes the loop through the impact crusher CR.5011, and the screen delivers KFS 20/35 to stock and the 0/20 stream to the downstream stockpile.

4.1 The validated inputs

The zone-1 inputs were validated first, before any machine sheet, per the step-by-step review protocol (client, 2026-08-11):

Input	Value	Provenance
Pivot feed rate	250 t/h wet (232.5 dry)	client-validated 2026-08-11; total-flow rule §1.1
Moisture	7 % wet basis	measured with the 2026-08-08 belt cut
Feed PSD	d50 32.3 mm, F80 180.6 mm	belt-cut 2026-08-08 (measured 32 / 180); H-FEED-1 fine-tail and H-FEED-2 top-size completions; Annex A
Bond work index	12.54 kWh/t	[ref.] — Q2 closed 2026-08-11
Settings	g = 60 mm / CSS = 30 mm / v = 30 m/s	client 2026-08-13 (g, CSS) and 2026-08-14 (v)
Mode / weather	1A / dry	default scenario

The three settings all sit at the yield-favorable end of their machine ranges — g and CSS at their maxima, v at its minimum — and each direction is engine-proven monotone (chapter 9, action 3). Zone 1.1 has no settings headroom left against the 0/20 balance: what remains is the quarry curve.

4.2 CR.5009 — toothed roll crusher

The roll crusher takes the full pivot flow at the adopted gap g = 60 mm ($x_{80} = g$, client validation 2026-08-08; RR slope n = 1.35). The engine reports:

Quantity	Value	Model
Throughput	250 t/h wet (232.5 dry)	—
F80 / P80	180.6 / 42.7 mm	M1
Specific energy W	0.312 kWh/t	M2

Net / installed power	72.5 / 96.6 kW	M2 (η_m 0.75)

> **Nip saturation (standing alert).** The measured feed F_{80} of 181 mm exceeds the CR.5009 maximum nip size of 150 mm — the engine raises this alert on every run of the measured curve. Top-size control at the primary crusher belongs to the same quarry conversation as the fines fraction (chapter 8), and the alert worsens on the coarser target curve (F_{80} ≈ 251 mm).

The gap g = 60 mm was adopted in the 0/20-balance optimization campaign of 2026-08-13: a coarser roll product preserves the 20–35 mm KFS band instead of grinding it into 0/20. The model applies the client-validated identity $x_{80} = g$ (2026-08-08) with the M1 product form of §3.1 (n = 1.35); the specification's ~116 kW chapter-9 power expectation is a known spec-side artefact of the back-fitted reference curve (Q2 closure) — the engine's 96.6 kW installed at the measured curve is the design number. The expert book's worked CR.5009 example uses an F80 of 120 mm where the belt cut measures 180.6 (66 vs 106 kW net at the book's settings) — one of the four points logged for the expert clarification note; the model always computes on the measured curve.

4.3 SR.5007 — double-deck screen 35/20 mm

The screen receives the crusher product plus the recirculating load — 301.8 t/h dry at loop equilibrium — and cuts at 35 and 20 mm with imperfection I = 0.15 [H] (M3). The VSMA sizing (M4, f_0 = 0.347 fitted, f_p = 1.2) gives the required areas:

Deck	Aperture (mm)	Basic capacity Q_b (t/h·m²)	Required area (m²)
Top deck	35	118.2	6.80
Bottom deck	20	84.5	7.15

Installed areas are open data slots (null) awaiting the design drawings; the design check reports "no limit" until they are filled. The partition of each deck follows the M3 efficiency curve at sharpness s = 13.52 (I = 0.15), roughly twice as sharp as a classic I = 0.15 screen under the disclosed convention nuance of §3.3 — a bias direction the Q3 sieve test will settle.

4.4 CR.5011 — impact crusher (HAZEMAG AP-S 1010) and the closed loop

The top-deck oversize (> 35 mm) recirculates through the impactor at the adopted settings CSS = 30 mm, v = 30 m/s — both at the yield-favorable end of their ranges (the monotonicity of each is engine-proven, chapter 9 action 3). The M5 chain at loop equilibrium:

Quantity	Value	Model
Ecs = v²/7200	0.125 kWh/t	M5 (exact identity)
t10 = A_j(1−e^(−b_j·Ecs))	5.71 %	M5 (A_j 60, b_j 0.8 — both [H])
RR slope n = (30/t10)^0.30	1.645	M5
F80 / P80	63.0 / 29.3 mm	M1
Net / installed power at loop load	16.1 / 21.5 kW	M2
Net / installed power at 90 t/h rating	21.0 / 28.0 kW	M2 (motor 132 kW installed)

The recirculating load at equilibrium is **69.3 t/h dry** (29.8 % of the zone feed). The capacity check runs in the rating's declared basis (audit fix, 2026-08-14): the vendor's 90 t/h real-capacity rating is wet, so the engine compares the wet loop load — 74.5 t/h — against it: **82.8 % of rating**, inside capacity. Mode 1B at 250 t/h would exceed it and is unusable at these settings — irrelevant, since the adopted balance needs no stock campaigns.

The loop is a fixed-point iteration on the recirculating stream (tolerance 10^{-6} on the size fractions, maximum circulating ratio 10, divergent configurations self-declared): the top-deck oversize feeds the impactor, whose product returns to the screen feed, and the engine iterates the pair to equilibrium. Action 5 of the verification program tightened the tolerance to 10^{-9} and confirmed the equilibrium unchanged — the loops are numerically clean.

4.5 KFS product

The 20/35 band reports to the KFS stockpile at **62.2 t/h wet** (57.8 dry). The real KFS PSD against its 30/55/15 envelope (three %-passing thresholds, interpretation validated 2026-08-08):

Envelope check	Limit	Realized	Verdict
Below cut (< 20 mm)	≤ 30 %	8.7 %	pass
In cut (20–35 mm)	≥ 55 %	82.7 %	pass
Above cut (> 35 mm)	≤ 15 %	8.6 %	pass

The full product curve on the presentation grid (P_{80} = 31.7 mm; meshes with 0 or 100 % passing omitted):

Mesh (mm)	10	12.5	15	20	25	31.5	35	40	50	63
Cumulative passing (%)	0.00	0.02	0.24	8.69	39.45	79.24	91.44	98.04	99.92	100.00

The product is envelope-compliant with comfortable margins on all three thresholds, and the compliance is robust: the Monte-Carlo band of chapter 9 found the envelope holding in 200 of 200 hypothesis draws (in-cut 77.9–92.7 %). The quality of KFS is not in question; only the quantity balance is.

The internal streams of the zone at loop equilibrium, dry solids:

Stream	Rate (t/h dry)	From → to
Pivot feed	232.5	pivot → CR.5009
Crusher product	232.5	CR.5009 → SR.5007 (joins the return)
Screen feed	301.8	= 232.5 + 69.3 recirculation
Top-deck oversize (> 35 mm)	69.3	SR.5007 → CR.5011 → back to SR.5007
KFS 20/35	57.8	SR.5007 → BC.5013 → KFS stock
0/20 undersize	174.7	SR.5007 → 0/20 stockpile (SP.5014)

4.6 The 0/20 stream and the zone balance

The bottom-deck undersize is the 0/20 stream at **187.8 t/h wet** (174.7 dry) — the feedstock of everything downstream and, through the reclaim, the physical carrier of the whole commercial balance. Its composition at the adopted settings drives the zone-1.2 split of chapter 5 (61.6 % coarser than 6 mm) and, two zones later, the fines/grits economics of chapter 6. The zone closes exactly:

Zone 1.1 stream	Wet (t/h)	Dry (t/h)
IN — pivot feed	250.0	232.5
OUT — KFS 20/35	62.2	57.8
OUT — 0/20 to stockpile	187.8	174.7
closure gap (dry, relative)	*$5.5{\times}10^{-8}$ — closed*	

Installed power engaged in the zone: 96.6 kW (CR.5009) + 21.5 kW (CR.5011 at loop load; 132 kW motor installed) — comminution is not the zone's binding constraint; the hours ceiling was (chapter 7), until the Saturday extension closed Q7.

How the settings got here — the zone-1 operating point is itself a decision trail, each step engine-quantified before adoption:

Epoch	g / CSS / v	Effect (engine, at adoption)
Spec-era published settings	40 / 20 / 45	KFS 51.35 t/h wet — the point the PFD's 80 t/h design rate is confronted with (RC7)
0/20-balance campaign (2026-08-13)	60 / 30 / 35	KFS yield 20.5 → 23.9 % (spec grid); envelope compliant; g and CSS at machine maxima
Last headroom (2026-08-14)	60 / 30 / **30**	landfill −9 800 t/y at that epoch (58.4 → 48.6 kt); envelope verified compliant
×2 grid rebase (2026-08-14)	60 / 30 / 30	KFS Yield 24.59 → **24.88 %** converged — this chapter's operating point

4.7 KFS Yield — realized against required

$$Y_{KFS} = \frac{62.20}{250} = 24.88\,\% \qquad Y_{KFS}^{req} = \frac{85\,000}{85\,000 + 242\,829} = 25.93\,\% \tag{4.1}$$

The realized yield sits 1.05 points below the requirement for zero landfill at the adopted annual plan — the engine alerts it on every run. The gap is structural, not a settings issue: g and CSS sit at their machine maxima and v at its minimum, all yield-favorable directions engine-proven; 45.5 % of the measured feed is already finer than 20 mm at the inlet and is fatally 0/20. The remaining lever is the quarry curve (chapter 8).

5 Zone 1.2 — complete calculation

The 100 t/h wet reclaim split at 6 mm into FeedLime and the AgLime loop — PFD REV18 topology, mode 2A photo at defaults.

Zone 1.2 reclaims the 0/20 stockpile at 100 t/h wet (93.0 dry) and was rebuilt to the NACO/Carmeuse PFD REV18 topology on 2026-08-12 (PFD authority ruling): FeedLime is the 6/20 oversize of the single-deck 6 mm screen SR.5105; the undersize feeds the AgLime finishing loop — open screen SR.5111, impact crusher CR.5113, and the closed-loop screen SR.5115, both screens cutting at 1.7 mm. All figures below are the engine's mode-2A dry photo at defaults.

The reclaim inputs carry two declared hypotheses, both arbitrated in the zone-1.2 question round of 2026-08-12: the reclaim PSD is the zone-1 photo's 0/20 curve at the validated settings (the stockpile is assumed well-mixed — the only consistent choice, since no downstream measurement will ever exist), and the reclaim moisture is the belt moisture of 7 %. The reclaim feeders BF.5101/BF.5102 are rated 100 t/h each (PFD REV18); routing units have no process influence.

5.1 SR.5105 — the 6 mm FeedLime cut

The reclaimed 0/20 (the zone-1.1 product curve of chapter 4) splits at 6 mm into **FeedLime 6/20 at 61.6 t/h wet** (57.3 dry, 61.6 % of the reclaim) and 38.4 t/h wet (35.7 dry) toward the AgLime loop. The VSMA required area is 3.01 m². The split is the mirror of the zone-1 product's fineness: the measured chain gives ~61/39, the inverse of the PFD design table's 42/58 — the design curve assumed a finer 0/20 than the measured chain produces (the same family as the KFS design-rate gap, register 2026-08-12; NACO reconciliation, chapter 10).

5.2 The AgLime loop — SR.5111 open, CR.5113 + SR.5115 closed

Machine	Load	Result
SR.5111 (1.7 mm, open)	35.7 t/h dry	undersize 7.9 t/h direct to AgLime; area 1.42 m²
CR.5113 (impactor, v 40 / CSS 1.0)	28.1 t/h dry	Ecs 0.222 kWh/t, t10 9.77 %, n 1.400; P80 0.95 mm; 65.5 / 87.4 kW net/installed
SR.5115 (1.7 mm, closed loop)	28.1 t/h dry	undersize joins AgLime; area 4.99 m²

The loop streams at equilibrium, dry solids: the SR.5105 undersize (35.7 t/h) reaches SR.5111, whose undersize — 7.9 t/h of natively fine material — reports directly to AgLime; the 27.8 t/h oversize joins the 0.3 t/h SR.5115 return at the crusher (28.1 t/h through CR.5113), and the crushed product passes SR.5115, 27.8 t/h joining AgLime and 0.3 t/h recycling. Two-stage closing nearly eliminates recirculation: **0.3 t/h** at equilibrium, where the pre-rebuild single-stage loop ran at 82 % circulating ratio. The loop conveyors' 60 t/h PFD rating (BC.5110/BC.5116) is checked on every run: the mode-2A loop feed of 28.1 t/h uses 47 % of it — ample.

5.3 AgLime product

The combined AgLime stream is **38.4 t/h wet**, of which **98.4 % passes 1.7 mm** — comfortably above the ≥ 95 % acceptance criterion adopted on 2026-08-12 (the project's first numeric AgLime specification; out-of-cut 1.6 % against the 5 % tolerance). The full product curve (P_{80} = 1.08 mm):

Mesh (mm)	0.063	0.1	0.125	0.25	0.5	1.0	1.5	1.7	2.0	2.8
Cumulative passing (%)	2.95	5.53	7.48	18.46	41.41	75.91	94.71	98.38	99.73	100.00

The wet balance closes exactly: 61.6 FeedLime + 38.4 AgLime = 100 t/h reclaim (dry: 57.3 + 35.7 = 93.0, engine closure gap 4×10^{-11} relative).

Zone 1.2 stream (mode 2A)	Wet (t/h)	Dry (t/h)
IN — 0/20 reclaim	100.0	93.0
OUT — FeedLime 6/20 to zone 1.3	61.6	57.3
OUT — AgLime 0/1.7 to stock	38.4	35.7
closure gap (dry, relative)	*4.0×10^{-11} — closed*	

5.4 The 2A/2C operating logic

Zone 1.2 runs on the strict demand rule: in mode 2A its hours are set by the zone-1.3 FeedLime demand, and AgLime is the co-product of those hours. Since the C1 adoption collapsed the FeedLime demand (2.8× fewer fines ground downstream), the 2A co-production alone leaves the AgLime market unserved; the client's rule of 2026-08-13 ("producing lime is never a problem — separate campaign"), wired on 2026-08-14 as `commercial_rules.aglime_2c_campaigns`, closes the gap with **dedicated mode-2C campaigns** in which the whole reclaim leaves as AgLime. Campaign hours count against the zone-1.2 ceiling and are alerted if capped. At the adopted plan (chapter 7) the split is 67 144 t from 2A co-production plus 67 856 t from 679 effective campaign hours — exactly the 135 kt market. Rain forces mode 2B (a mass identity, FeedLime = whole reclaim); at the adopted plan the rain complement is zero hours (dry/rain season split 75/25 as parameters, never imposed by the engine).

The engine's mode-2C photo at the full 100 t/h wet reclaim: AgLime 100 t/h wet, 99.4 % below 1.7 mm — comfortably compliant, the campaign product being finer than the 2A co-product because everything passes the crushing loop. Two consequences are alerted and belong on the design table:

> **2C campaign loading (standing alerts).** At the full 100 t/h reclaim the 2C photo drives the AgLime loop at 86.0 t/h dry through CR.5113 — the engine flags the loop feed of 100 t/h wet against the 60 t/h conveyor rating (BC.5110/BC.5116, PFD REV18), and the computed CR.5113 requirement rises to 261 kW net (348 kW installed basis) against the 87 kW of mode 2A. Either the campaign reclaim rate is derated to the loop rating — lengthening the 2C hours accordingly, ceiling margin is ample — or the loop conveying and motor are sized for campaign duty. To reconcile with the engineering contractor; the annual tonnages of chapter 7 are unaffected (hours scale inversely with rate).

Under rain the 1.7 mm cut is physically impossible on the outdoor screens and the engine replaces mode 2A by 2B, alerted (the rule is a data parameter, disengageable). The rain photo at defaults: the whole 100 t/h wet reclaim leaves as FeedLime — SR.5105 inactive, no AgLime production — and zone 1.3 then runs on straight 0/20 (grits 16.62 t/h, fines 13.15 t/h in mode G: nearly unchanged, the 6/20 insulation again). Because the adopted plan needs zero rain-season zone-1.2 hours, rain is a scheduling constraint (AgLime hours belong to the dry season), not a capacity one; the rain-mode screen derating (basic-capacity factor 0.75 [H]) is carried in M4 for the day it matters.

5.5 Stockpiles and the seasonal watch

The PFD yard capacities are recorded as data (KFS fines 0/20: 10 000 t at SP.5014; KFS 20/35: 5 000 t; Feed 0-6/20: 6 000 t at SP.5109; AG 0/1.7: 28 000 t at SP.5118) but not yet enforced by the engine. Two watches stand: the FeedLime interstage stock is a weekly buffer (the 6 000 t capacity is ample for the ~40 clock-hours-per-week demand rhythm, per the 2026-08-12 priority ruling), and the earlier planning epochs found a seasonal FeedLime swing larger than the yard when rain-mode hours concentrate — a sizing check to repeat when the operating calendar firms up. At the adopted plan the 0/20 stockpile nets to zero over the year by construction (the landfill line absorbs the excess), so no yard-growth alert remains.

6 Zone 1.3 — the redesign

Why the as-built circuit was condemned, what C1 is, and the two operating modes that serve both the grits and the fines markets.

6.1 As-built diagnosis — the case for the redesign

The as-built zone 1.3 (dryer DY.03 → classification screen SN.21 → toothed roller mill ML.26 in closed circuit) was condemned on three engine-proven counts. Re-run at today's HEAD (as-built variant selectable, converged grid, full 32.1 t/h wet feed), the circuit grinds **21.0 t/h of fines for 7.8 t/h of grits — a fines/grits ratio of 2.7** (2.83 on the diagnosis-era grid, register 2026-08-14) where the design basis demands ≤ 1.25 firm. Its grits **fail the D6 envelope: 15.4 % below 2 mm** against the 15 % limit (and 5.3 % above 4 mm against 5 %). And the 80 kt/y extension regime is **infeasible**: at 7.8 t/h of grits it needs over 10 000 effective hours against the 6 000 h ceiling (register: 10 275 h). The panel diagnosis attributed the ratio, engine-backed: ML.26 compression spread (n_{comp} 0.86) 67 %, bed attrition (S_{att} 0.171) 28 %, sliver-band leak 5 % — and the feed is blameless. Counter-intuitively, the closed loop and screen imperfection are *not* ratio destroyers (perfect screens worsen the ratio to 3.65); an 810-run settings sweep floors the existing machine at 2.90. **It is a machine problem** — confirmed by candidate C3, which kept the toothed rolls in a fixed topology and still returned 2.90.

The condemnation in one table — the as-built circuit and the adopted C1 mode-G photo at the identical duty (full 32.1 t/h wet dryer feed, both engine runs at today's HEAD):

Quantity	As-built (SN.21 + ML.26)	C1 mode G (RC.1/RC.2 + SC.A/SC.B)
FeedLime grits 2/4	7.76 t/h (26.0 % of dried)	**16.22 t/h** (54.3 %)
FeedLime fines 0/1.5	21.03 t/h	13.57 t/h
Fines/grits ratio (D3: ≤ 1.25 firm, ≤ 1.0 objective)	2.71	**0.84**
Grits < 2 mm (D6: ≤ 15 %)	15.4 % — **FAIL**	14.2 % — pass
Grits > 4 mm (D6: ≤ 5 %)	5.3 % — **FAIL**	3.4 % — pass
UltraFin (not certified)	1.07 t/h	0.07 t/h
Recirculating load	39.6 t/h	62.4 t/h
Effective hours for 80 kt/y grits (D5 ceiling 6 000)	> 10 300 — **infeasible**	~4 800 (with the BC.22 retrofit)

The panel's derived purchase-grade machine requirement: the final-stage crusher must place at least 40 % of each crushed tonne in 2–4 mm with at most 29 % below 1.5 mm per pass — a Rosin–Rammler slope n ≥ 1.6 at $x_{80} \sim$ 3.5–4 mm. Smooth double rolls at reduction ratio ≤ 2–2.5 are the only credible machine class [H — the vendor gradation test on WANKOE 6/20 is trial trigger #1]. A structural robustness gift came free: the FeedLime 6/20 double cut leaves the zone-1.3 feed with no native 2–4 band and no fines, so both D4 feed qualities give identical zone-1.3 results — the redesign is insulated from the quarry question.

6.2 Candidate circuits — panel round 1

Candidate	Concept	Fines/grits ratio	Verdict
C1	two-stage smooth double rolls + grading screens with immediate 2–4 relief	0.98–1.05 (worst-sensitivity 1.13)	**BUILD recommendation — adopted**; 5 units; 80 kt at 88.6–91.5 % of ceiling
C2	three-stage extra-gentle (reduction ~1.7 per pass)	0.84 (0.99 at n = 1.6)	escalation reserve if the vendor test returns n < 1.7; 7 units
C3	topology fix, toothed rolls kept	2.90	REJECTED — proves the machine, not the circuit, is the root cause
C4	quad-roll compact (2 units)	1.10, but 1.30 at S_att = 0.10	crosses the firm limit on a plausible coefficient — capex fallback only
C5	low-speed impactor stage 1	1.09, but 1.30 at honest n = 1.0	marginal; blowbar operating cost at 7 500 h

All ratios are engine results on the D4 design feed across the honest [H] coefficient band (panel round 1, 2026-08-14). C1 was validated as lead candidate the same day, wired into the engine as a selectable variant, and then formally adopted as the reference configuration after the arbitrations below.

6.3 The C1 circuit

Candidate C1 — validated as lead candidate and then formally adopted as the reference configuration on 2026-08-14 — replaces the mill with **two-stage smooth double rolls with immediate product relief**:

Unit	Duty	Purchase specification
RC.1	stage-1 smooth double roll, gap 8 mm	32 t/h (client 2026-08-14; closes the D4 quarry-feed finding — the 29 t/h panel sizing was exceeded at 29.7 t/h on the target curve)
RC.2	stage-2 smooth double roll, gap 3.4 mm (mode G) / 1.5 mm (mode F)	TWO units × 22 t/h; minimum gap 1.5 mm — vendor to confirm
SC.A	double-deck RECYCLE screen 8 / 3.75 mm	required areas 4.8 / 5.6 m² (worst mode)
SC.B	double-deck PRODUCT screen 2 / 1.5 mm	required areas 5.2 / 4.5 m² (worst mode)
Sliver diverter	SC.B deck-2 oversize (1.5/2 band)	two-position: REGRIND through RC.2 (adopted, option B) / extract as Sliver 1.5/2
DV.GF grits diverter	SC.B deck-1 oversize (2/3.75 band)	two-position: grits product chain (mode G) / RC.2 regrind circuit (mode F)

The screen arrangement is the client's "2+2" of 2026-08-14: SC.A carries the recycle cuts (8 mm back to RC.1, 3.75 mm back to RC.2), SC.B carries the two quality-critical product cuts — the 2.0 mm deck protecting the D6 < 2 mm limit and the 1.5 mm deck protecting the 0/1.5 fines specification and its redirect eligibility — each in the favorable deck position. A linking conveyor carries the SC.A undersize (0/3.75, ~41 t/h in mode G) to SC.B. The sliver arbitration (option B) reground the 1.5/2 no-man's-band through RC.2 after an engine counterfactual showed regrind *wins* with smooth rolls (S_{att} 0.06): ratio 0.79 versus 1.05 extracted, with D6 still met — the panel's "never regrind" rule had been calibrated on the toothed mill. The dryer DY.03 and the optional SP.36/CL.38 UltraFin block are unchanged (D2).

6.4 The adoption sequence and its commercial cascade

The path from candidate to reference configuration ran through five same-day arbitrations (2026-08-14, register rows in chapter 2), and the honest record includes the commercial shock the adoption caused. First the sliver arbitration (regrind, ratio 0.79) and the 2+2 screen arrangement; then a "base scenario" with a single RC.2 unit and the dryer throttled so the unit runs exactly at its 22 t/h loop limit — re-bisected on the converged grid to a feed of 21.29 t/h wet, grits 10.76 t/h, single-unit maximum capacity 64 554 t/y. When C1 became the default, the engine exposed the **cascade of making 2.8× fewer fines**: fines production collapsed below the 60 kt market, the fines redirect dried up, the strict demand rule capped zone 1.2 at the collapsed FeedLime demand, AgLime fell to loop-only 49 kt against its 135 kt market, and the 0/20 landfill exploded from 18.5 to 144.2 kt/y. The client then pulled the levers in order: the dedicated 2C campaigns (landfill 144.2 → 58.4), the impactor at v = 30 (→ 48.6), the converged grid rebase (→ 42.5), and finally the fines objective with the two-mode design below (→ **13.8 kt/y**), leaving the quarry curve of chapter 8 to close the rest. Every step of that history is a dated register row with its engine numbers.

6.5 The two operating modes — engine results

The fines objective of 2026-08-14 (60 kt/y must be *served*, not merely capped) added the grits diverter and the second RC.2 unit, and made zone 1.3 a two-mode plant. Both photos below are engine runs at the adopted defaults:

Quantity	Mode G — grits	Mode F — fines campaign
Feed (wet)	32.10 t/h (full dryer limit)	25.05 t/h
Dryer outlet (wet / dry)	30.00 / 29.85 t/h	23.41 / 23.30 t/h
FeedLime grits 2/4	**16.22 t/h** (54.3 % of dried)	0 (diverted)
FeedLime fines 0/1.5	13.57 t/h	**23.19 t/h**
UltraFin (not certified)	0.07 t/h	0.10 t/h
RC.1 load / spec	29.2 / 32 t/h	22.8 / 32 t/h
RC.2 load / installed (gap)	33.2 / 44 t/h (3.4 mm)	**44.0 / 44 t/h** (1.5 mm)
Recirculating load	62.4 t/h	66.8 t/h
Grits QC: < 2 mm / > 4 mm	14.2 % / 3.4 % — D6 met	—
Fines < 1.7 mm (redirect eligibility ≥ 95 %)	96.6 %	98.9 % (94.4 % < 1.5)
RC.1 + RC.2 installed power	17.4 + 37.9 kW	13.6 + 48.5 kW
Dryer burner / vapor	3 718 kW / 2.10 t/h	2 901 kW / 1.64 t/h

Mode F runs both RC.2 units exactly at capacity — 25.05 t/h wet is the bisected feed at which the constraint binds. The mode-G point runs the dryer at its full 32.1 t/h wet limit, maximizing grits co-production per hour; the diverter DV.GF switches the plant between the two order books without touching a single machine.

The internal streams of the C1 circuit at both operating points, dry solids:

Stream	Mode G (t/h dry)	Mode F (t/h dry)
Dryer outlet → RC.1 circuit	29.85	23.30
RC.1 load (fresh + 8 mm recycle)	29.2	22.8
SC.A feed	92.2	90.1
Linking conveyor SC.A → SC.B (0/3.75)	40.8	50.0
RC.2 load (3.75 recycle + sliver regrind [+ 2/3.75 in mode F])	33.2	44.0
Total recirculating load	62.4	66.8
Products (grits + fines + UltraFin)	16.22 + 13.57 + 0.07	0 + 23.19 + 0.10

6.6 Product size distributions

The three product curves at the adopted operating points (presentation grid, meshes at 0 or 100 % omitted; all dry-basis zone-1.3 products at 0.5 % moisture):

Mesh (mm)	0.063	0.1	0.25	0.5	1.0	1.5	1.7	2.0	2.8	4.0	5.0
Grits 2/4, mode G (P80 3.32 mm)	—	—	—	—	—	0.11	1.79	**14.18**	56.06	**96.63**	99.80
Fines 0/1.5, mode G (P80 1.41 mm)	0.16	0.92	5.85	18.51	51.32	86.13	**96.61**	99.95	100.00	—	—
Fines 0/1.5, mode F (P80 1.31 mm)	0.15	0.86	5.72	18.86	55.08	94.41	**98.88**	99.96	100.00	—	—

The bold cells are the specification-critical readings: grits 14.18 % below 2 mm (D6 limit 15 %) and 96.63 % below 4 mm (i.e. 3.37 % oversize against the 5 % limit); fines 96.6 / 98.9 % below 1.7 mm against the ≥ 95 % redirect-eligibility threshold. The mode-F fines are markedly tighter than mode G's — the 1.5 mm regrind gap converts the 1.5/2 sliver band into sellable fines instead of letting it dilute the top of the curve.

Turndown and degraded operation are covered by the same diverters. Phase-1 stopgap, on record from the base-scenario study: with the sliver diverter on "extract", a single RC.2 unit carries the full dryer (grits 14.6 t/h, 87.5 kt/y maximum, unit at 98.5 % load) at the cost of a 4 t/h Sliver 1.5/2 by-product — one unit down does not stop the zone. And mode G itself can run one unit at throttled feed or two at full dryer; the annual plan is indifferent to how the operator schedules the G/F campaign blocks within the 60.2 % utilization.

6.7 The mode-F gap physics finding

Mode F is only feasible because the purchase specification extends the RC.2 gap range down to 1.5 mm. The engine finding that forced it: at the machine-class minimum gap of 2.8 mm, the diverted 2/3.75 band can never be nipped — everything between the 2.0 mm screen cut and the 2.8 mm gap bypasses the rolls unbroken and returns to the screen, forever. Re-run at today's HEAD at the adopted mode-F feed, the RC.2 load diverges to **172.8 t/h against 44 t/h installed** (the register's design-study figure was 147 t/h at its study feed, 2026-08-14); at the 1.5 mm gap the same circuit settles at 44.0 t/h. The 1.5 mm capability is therefore a **hard purchase requirement**, flagged vendor-to-confirm, and belongs to the same gradation test as the [H] coefficients.

> **D6 margin watch.** Under sliver regrind the mode-G grits carry 14.2 % below 2 mm against the 15 % limit — a margin of 0.8 point that rests on the [H] smooth-roll coefficients (n_{comp} 1.8, S_{att} 0.06). The vendor gradation test must confirm the margin before the purchase is closed; the diverter keeps the extract fallback reversible in operation.

6.8 Conveying consequences

At the adopted operating points the BC.22 grits conveyor stays within its 15 t/h PFD rating in the 40 kt plan; the full-dryer regrind extension case (grits 16.7 t/h) exceeds it and carries a flagged retrofit to ~20 t/h. The BE.40 fines conveyor, overloaded by the as-built circuit (21.0 t/h), returns within its rating under C1. Both are chapter-10 items.

6.9 Purchase specifications — consolidated

Item	Specification	Basis	Open point
RC.1 smooth double roll	32 t/h, gap range around 8 mm	dry solids (engine-derived loop load; client 2026-08-14)	gradation test fixes n_comp / S_att; capacity at gap
RC.2 smooth double rolls	**two units** × 22 t/h; gap range 3.4 mm down to **1.5 mm**	dry solids; mode-F capability is a hard requirement (§6.7)	1.5 mm gap capability vendor-to-confirm
SC.A recycle screen	double deck 8 / 3.75 mm; required 4.8 / 5.6 m²	VSMA (±20–30 %); worst mode F	installed areas and motors to size — recommend sizing for the 80 kt extension (D1)
SC.B product screen	double deck 2.0 / 1.5 mm; required 5.2 / 4.5 m²; sliver diverter + DV.GF grits diverter	VSMA; both quality-critical decks in favorable position	1.5 mm deck efficiency at scale
Linking conveyor SC.A → SC.B	≥ 50 t/h of 0/3.75 (mode-F load)	engine stream table §6.5	rating to fix at design

BC.22 grits conveyor	15 t/h adequate at 40 kt; ~20 t/h for the extension	PFD rating vs engine loads	retrofit deferred with the extension provision

7 Annual plan and consolidated balance

Hours follow the targets, never the reverse — the full-year balance at today's feed and at the quarry target.

The planning engine (`run_required_hours`) computes the hours each zone needs to land its targets, respecting the shift-regime ceilings as capacity limits (client rule 2026-08-08). Two engine executions are consolidated here: the default parameter set (measured 2026-08-08 feed, AgLime market 135 kt) and the quarry-target variant (target curve of chapter 8, AgLime baseline 108 kt).

The shift frames behind the ceilings:

Zone	Regime	Clock ceiling (h/y)	Availability	Effective ceiling (h/y)
1.1	1 shift × 6 d/7 — the Saturday extension (Q7, 2026-08-13)	2 400	80 %	1 920
1.2	3 × 8 h continuous	7 500	80 %	6 000
1.3	3 × 8 h continuous (D5)	7 500	80 %	6 000

Time bases other than annual (monthly, weekly, daily) are parameters of the same engine; the annual basis is used throughout this report. Utilization percentages below are quoted against the clock ceilings, as the engine reports them.

7.1 Hours and mode splits

Zone	Measured feed	At quarry target	Ceiling (clock)
1.1 — effective / clock h	1 367 / 1 708 (71.2 %)	1 203 / 1 504 (62.7 %)	2 400 (Saturday regime)
1.2 — effective / clock h	2 428 / 3 035 (40.5 %)	2 158 / 2 698 (36.0 %)	7 500
1.3 — effective / clock h	3 611 / 4 513 (60.2 %)	3 612 / 4 515 (60.2 %)	7 500
Zone-1.3 split: mode G + mode F (eff.)	2 466 + 1 144 h	2 460 + 1 152 h	—
Zone-1.2 split: 2A + 2C campaigns (eff.)	1 750 + 679 h	1 576 + 582 h	—

All zones are feasible with margin; zone 1.1 is KFS-driven in both cases. Zone 1.3 is insensitive to the quarry curve by construction — the 6/20 double cut insulates it (the panel's structural-robustness finding). In the weekly translation the kiln contract reads 1 634.6 t of KFS per week (85 000/52, continuous kiln; register 2026-08-12): zone 1.1 delivers it in about 26 effective hours per week within its 6-day, one-shift Saturday regime, keeping roughly one shift per week of buffer — the slack Q7 was closed to create. The utilization percentages above are the design margin statement: no zone above 72 %, no seasonal squeeze, and the two zone-1.3 modes schedulable as campaigns at the operator's convenience.

7.2 Consolidated zone-by-zone balance (annual, wet tonnes)

Stream	Measured feed (t/y)	At quarry target (t/y)
Zone 1.1 IN — quarry feed at 250 t/h wet	341 650	300 825
Zone 1.1 OUT — KFS 20/35 to stock	85 000	85 000
Zone 1.1 OUT — 0/20 to stockpile	256 645	215 830
closure gap	*−0.1 (< 0.001 %)*	*−0.4 (< 0.001 %)*
0/20 reclaimed downstream	242 829	215 830
0/20 to LANDFILL (net loss)	**13 816**	**0**
Zone 1.2 IN — reclaim at 100 t/h wet	242 830	215 830
Zone 1.2 OUT — AgLime sold	135 000	108 000
Zone 1.2 OUT — FeedLime 6/20 to zone 1.3	107 829	107 830
closure gap	*+1.0 (< 0.001 %)*	*0.0*
Zone 1.3 IN — FeedLime (mode G + mode F)	107 829	107 830
Zone 1.3 OUT — FeedLime grits 2/4 (dry)	40 000	40 000
Zone 1.3 OUT — FeedLime fines 0/1.5 (dry)	60 000	60 000
Zone 1.3 OUT — UltraFin (dry, not certified)	284	285
Zone 1.3 OUT — dryer vapor	7 044	7 044
Zone 1.3 OUT — residual product moisture (0.5 %)	504	504
closure gap	*−3.1 (0.003 %)*	*−3.1 (0.003 %)*

Every closure gap is below 0.003 % — presentation rounding of engine streams whose per-run balances close to better than 2×10^{-6} relative (dry solids and water asserted on every run). The FeedLime interstage stock is balanced (produced = consumed) in both cases; the 0/20 stockpile nets to zero, the landfill line absorbing the excess at the measured feed.

7.3 Sales

Sales line	Measured feed (t/y)	At quarry target (t/y)
KFS 20/35 (firm)	85 000	85 000
FeedLime grits 2/4 (firm)	40 000	40 000
FeedLime fines sold as fines	60 000	60 000
Fines surplus redirected to the AgLime sales channel	0	0
AgLime from loop (2A co-production)	67 144	49 814
AgLime from dedicated 2C campaigns	67 856	58 186
AgLime total sold / market cap	135 000 / 135 000	108 000 / 108 000 (+27 000 flex reserve)
UltraFin (market to develop, Φ not certified)	284	285

0/20 to landfill (net loss)	**13 816**	**0**

The two-mode zone 1.3 serves all four markets exactly in both cases; the fines redirect is zero because mode F closes the fines market directly. The KFS Yield reads 24.88 % realized against 25.93 % required at the measured feed, and a self-consistent 28.26 / 28.26 % at the target — the balance's whole story in one indicator pair.

7.4 How the landfill line got here

The 13 816 t/y residual is the end of a documented descent, each step a dated decision with its engine number — the honest measure of how much design work the balance absorbed:

Configuration epoch (register date)	0/20 to landfill (t/y)
C1 adopted, strict demand rule, no campaigns (2026-08-14)	144 200
+ dedicated AgLime 2C campaigns (2026-08-14)	58 400
+ impactor v = 30 m/s (2026-08-14)	48 600
+ converged ×2 grid rebase (2026-08-14)	42 500
+ fines objective, two-mode zone 1.3 (2026-08-14) — **this report's plan**	**13 816**
+ quarry target curve 40.1 % < 20 mm (chapter 8)	**0**

The remaining internal lever is commercial, not technical: grits sales beyond the firm 40 kt toward the chain's capacity absorb roughly 1.4 t of landfill per extra grits tonne (register arithmetic, 2026-08-13/14). The structural lever is the quarry curve — and action 4 of chapter 9 proved it indispensable: no plausible combination of model hypotheses closes the landfill gap at today's feed.

7.5 UltraFin and the extension case

UltraFin extraction is optional (D7) and marginal at the C1 configuration: 284 t/y against a 4 000 t/y target, because the smooth-roll circuit — that is its virtue — barely produces sub-65 µm material, and the figure remains NOT CERTIFIED until Φ at the cut is measured (§3.8). The 80 kt/y grits extension (D1's sizing case) is served by design, not by the plan: the machine park handles it (mode-G full-dryer regrind, ~4 800 effective hours at 80 kt, 80 % of the D5 ceiling) once the BC.22 conveyor is retrofitted, and the layout keeps the civil reserve; the 40 kt plan simply runs the same park fewer hours. What the extension also needs — more 0/20 demand upstream feeding more zone-1 hours — was quantified when the 80 kt case was recorded as an extension study (register 2026-08-13) and is to reopen with the engineering contractor if the market firms up.

7.6 Standing engine alerts at the adopted plan

The engine's own alert list, reported verbatim as the honest margin statement of the plan (measured feed; the same run that produced the tables above):

Alert (engine text, condensed)	Disposition
AgLime 2C campaigns: 679 effective hours produce 67 856 t to complete the market — consumes 67 856 t of 0/20	by design (§5.4); campaign loading alerts to reconcile with the contractor

0/20 excess 13 816 t/y to LANDFILL — net financial loss; reduce via grits sales, settings or KFS yield	the chapter-8 quarry target closes it to zero
KFS Yield 24.9 % < 25.9 % required for zero landfill	same — the live monitor of the quarry protocol
CR.5009: feed F80 181 mm > max nip size 150 mm — saturation	top-size control at the primary (§4.2, chapter 8 caution)
SP.36: Φ(< cut) not measured — UltraFin NOT CERTIFIED	measurement trigger (chapter 10)

8 Quarry works specification

The adopted inlet-PSD target: zero landfill with a 20 % AgLime-market margin.

8.1 Definition of the target

The firm and objective sales (KFS 85 kt, grits 40 kt, fines 60 kt) fix the downstream demand for crushed 0/20; the AgLime market absorbs the remainder through dedicated campaigns. The client's margin rule holds **20 % of the AgLime market open** as operational flexibility: the target curve is calibrated so the annual balance lands at

$$A_{baseline} = 0.8\, A_{cap} = 108\,000 \text{ t/y}, \qquad L_{0/20} = 0 \tag{8.1}$$

The curve is a shape-preserving size rescale of the measured 2026-08-08 belt-cut, $P_{tgt}(x) = P_{meas}(x/k)$, with k found by bisection on the full two-mode annual plan. The result is $k = 1.426$, i.e. a feed whose every characteristic size is 43 % coarser (D_{50}: 32 → 46 mm), delivering

$$\boxed{P_{tgt}(20\text{ mm}) = 40.1\ \%} \qquad (\text{measured today: } 45.5\ \%) \tag{8.2}$$

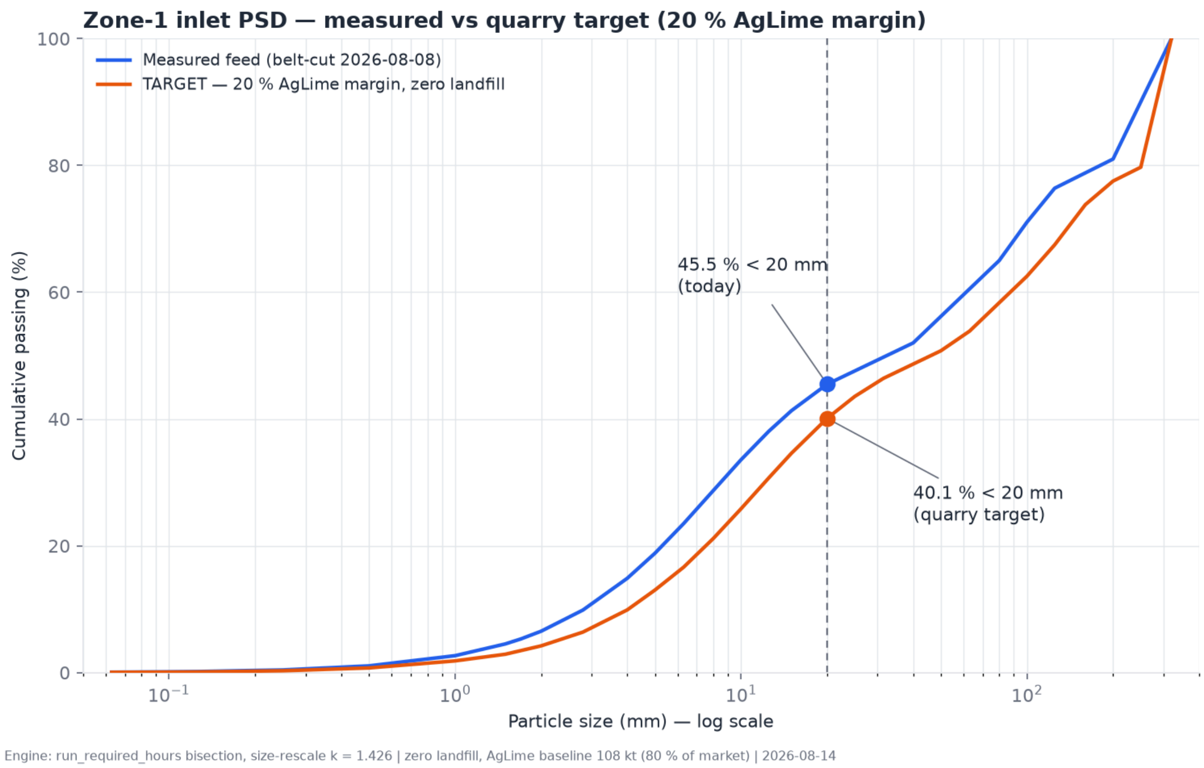

Fig. 8.1 — Zone-1 inlet PSD, semi-log: measured curve vs adopted target. The 20 mm control point carries the whole specification.

8.2 What the target buys

Annual quantity	Today's feed	At target
Quarry extraction for the same firm sales	341 650 t	300 825 t (−40 825)
0/20 to landfill (net loss)	13 816 t	**0**
AgLime baseline / flex reserve	135 000 / 0 t	108 000 / **27 000 t**
KFS Yield realized vs required	24.88 / 25.93 %	28.26 / 28.26 % — self-consistent
Zone utilizations 1.1 / 1.2 / 1.3	71.2 / 40.5 / 60.2 %	62.7 / 36.0 / 60.2 %

The explicit cost of the margin: 27 000 t/y of AgLime sales are not planned in the baseline. They are recoverable at will — whenever the quarry runs finer than target, the excess 0/20 is absorbed by pushing the 2C campaigns toward the full 135 kt market instead of being landfilled. The margin is bought with optional sales, not with waste.

8.3 Control protocol

Step	Rule
Measurement point	belt-cut PSD at the PIVOT (downstream of the primary crusher), standard project protocol
Control point	cumulative passing at **20 mm**
≤ 40.1 %	on target or coarser — full flexibility available
40.1 – 45.5 %	workable — AgLime campaigns absorb the excess; the flex reserve shrinks accordingly
> 45.5 %	worse than today — landfill exposure returns; escalate to the quarry
Live monitor	the engine's KFS Yield alert: each point of gap ≈ 12 kt/y of landfill exposure

> **Top-size caution.** Coarser than target is safe for the balance but not for the machines: the measured feed F_{80} (181 mm) already exceeds the CR.5009 maximum nip size (150 mm — standing engine alert). Top-size control at the primary crusher matters as much as the fines fraction; both belong to the same quarry conversation.

9 Verification and confidence

Six internal actions, a doubt budget spent down from 26 points to 6, and the honest statement of what stays external.

On the client's order ("maximum rigor and honesty — internal actions only, one at a time, quantified gains"), the zone-1.1 calculation chain — the root of every downstream figure — went through a six-action confidence program on 2026-08-14. The starting meta-score of 74 % was decomposed into a doubt budget; each action attacked one component, and the score moved only when the component demonstrably closed. Full reports: `docs/design/zone11-verification/`, all replayable (Annex C).

The initial decomposition of the 26 doubt points, each component with its designed attack: implementation errors (5 pts — could the code diverge from the intended formulas?), formula transcription (4 pts — were the right formulas chosen and copied?), physical behavior (3 pts — does the model move in the right directions?), numerics (2 pts — grid and loop convergence), the naked [H] point figure (8 pts — a single yield number resting on unmeasured coefficients), and feed representativeness (4 pts — one belt-cut carries the denominator). The first three and the numerics are internally closable; the last two are only partially so, which is why 94 % is a ceiling and not a stopping point.

9.1 Action 1 — blind independent recalculation (74 → 79)

Mechanism. A verification agent, forbidden to read the source code, the tests and the engine reference extract, rebuilt the entire zone-1.1 chain in pure standard-library Python from the documentation and the data file alone. **Verdict: PASS, full concordance** — 25 scalar values (powers, F80/P80, areas, recirculation, streams) and the full 29-point KFS curve match with worst deviation below 5×10^{-3} (display rounding); a replay at the old published settings reproduced the issued zone-1 calculation document exactly, proving the implemented conventions at two operating points. All 16 formula-detail assumptions the agent made alone landed on the engine's documented convention. The implementation-error component is closed. An excerpt of the concordance (action-era reference settings, spec grid):

Quantity	Engine	Independent	Deviation
KFS Yield (wet/wet)	24.59 %	24.59 %	0
KFS product	61.478 t/h wet	61.478 t/h wet	0
0/20 stream	175.325 t/h dry	175.325 t/h dry	0
Recirculation (CR.5011 load)	69.261 t/h	69.261 t/h	0
KFS cumulative curve, 29 points	—	—	0.00 at every point
CR.5009 W / P_net / P_inst / F80 / P80	0.312 / 72.468 / 96.624 / 180.575 / 42.709	idem (2-decimal rounding)	$< 5\times10^{-3}$
CR.5011 Ecs / t10 / n / P_net / F80 / P80	0.125 / 5.71 / 1.645 / 16.163 / 63.022 / 29.277	idem	$< 3\times10^{-3}$
SR.5007 VSMA areas	6.80 / 7.18 m²	6.80 / 7.18 m²	0

9.2 Action 2 — literature anchors (79 → 83)

Mechanism. Each model confronted with its source formula, every expected value hand arithmetic with the source cited, in a permanent test module. **Verdict: 7/7 pass** (table in §3.9), including one incident where the engine corrected the checker's own arithmetic, and the quantified disclosure of the M3 convention nuance (classic imperfection ≈ 0.081 at input I = 0.15). The transcription component is closed.

9.3 Action 3 — physical invariants (83 → 86)

Mechanism. The full stress campaign re-run at HEAD — 580 runs including the adopted C1 circuit: 0 failures on balances, curve monotonicity, finiteness and non-negativity — plus a new permanent module proving nine direction laws: coarser feed, wider gap, wider CSS and slower impactor all raise KFS Yield strictly; imperfection degrades in-cut monotonically; the M1/M2/M3/M6 limit behaviors are exact. **Verdict: PASS.** Every optimization decision of the project (g max, CSS max, v min, quarry coarsening) now rests on an engine-proven monotonicity, not a plausible narrative.

9.4 Action 4 — Monte-Carlo uncertainty band (86 → 89)

Mechanism. 200 seeded draws over the five uncertain inputs, each range traceable to a register row: CR.5009 slope n [1.05; 1.65], A_j [50; 69], b_j [0.6; 1.3], I [0.10; 0.20], feed size-rescale [0.93; 1.07] — uniform and independent, because no better information exists before the vendor tests. **Verdict:**

KFS Yield (action-era reference settings)	Value
P50 (median)	**24.3 %**
P10 – P90	22.4 – 26.0 %
Full range (200 draws)	21.6 – 27.1 %
Implied 0/20 landfill, P10 / P50 / P90 (t/y)	30 474 / 52 715 / 83 234
KFS envelope compliance	**200 / 200 draws** (in-cut 77.9 – 92.7 %)
Labeled literature-tail scenario outside the band (b = 3.0, A = 69)	yield 21.8 %, landfill ~93 kt/y — the drop-weight test decides

Product quality is hypothesis-robust; only the quantity balance moves. The conclusion that survives every draw: the zero-landfill requirement lies above the P90 and even above the best of the 200 draws — **the quarry lever is indispensable under every plausible hypothesis combination**, and landfill stays strictly positive in all 200 draws. The project's central recommendation is hypothesis-robust, not an artifact of one parameter choice. The remaining points of this component are the [H] values themselves and the feed representativeness — external by nature.

9.5 Action 5 — numerical robustness (89 → 91)

Mechanism. Grid refinement ×2/×4 by geometric midpoints and loop tolerance tightened to 10^{-9}, physics untouched. **Verdict: a real finding.** The fixed-point loops are clean, but the 29-mesh spec sieve grid carried a quantified discretization bias: KFS Yield underestimated ~0.32 pt (converged 24.91 %), in-cut overestimated ~3.3 pts (converged 82.1 %, still compliant), C1 grits overestimated ~0.32 t/h. Smaller than the Monte-Carlo band — no conclusion flips — but found and fixed internally: the client adopted the ×2 grid as the engine default and all figures were rebased (this report's 24.88 % is the converged value).

9.6 Action 6 — adversarial formula review (91 → 94)

Mechanism. Two independent adversarial audits (units/conventions lens; formula-fidelity lens), every claim numerically probed, every finding adjudicated. **Verdict: zero sign, inversion or exponent errors in M1–M8.** Three code fixes landed: the M7 capped-regime bypass pinned at the gap (material wider than the roll gap no longer survives a capped pass); capacity checks compare in each rating's declared basis (closing a ~7.5 % wet/dry blind spot on CR.5011); and M6 reports zero duty on a no-drying pass (was a ~1.2 MW phantom). The converged default results were unchanged by all three code fixes. The adjudication in full:

Category	Count	Content
Code fixes	3	M7 capped-regime bypass at the gap; per-machine capacity_basis (wet/dry); M6 zero duty on no-drying
Doc/data fixes	6	M3 convention note with conversion formulas; M1 truncation anchor disclosed; Phi measure-at-the-cut trap defused; unit labels (A_j %, densities kg/m³); RC.2 flow-split wording; zone_1_1 weather argument
Disclosed, no change	6	VSMA on dry tonnage (absorbed by the f0 fit); loop criterion naming; M1 truncation between meshes (< 0.5 %); attrition-fines truncation; dryer outlet sensible heat (~0.7 %); dual sub-mesh convention (< 1 %)
Verified clean by probe	—	mm/µm, wet/dry at every model boundary, %/fraction, kWh/t → kW (η applied once), M6 thermal chain, M8 Lapple, per-hour/per-year, RR conventions, water bookkeeping, C1 regrind composition

9.7 The doubt budget

Doubt component	Points	Closed by	Meta-score after
Implementation errors	5	Action 1 — blind recalculation	79 %
Formula transcription	4	Action 2 — literature anchors	83 %
Physical behavior	3	Action 3 — invariants + stress	86 %
Naked [H] point figure	3 of 8	Action 4 — Monte-Carlo band (internal share)	89 %
Numerics	2	Action 5 + ×2 grid adoption	91 %
Transcription residual	3	Action 6 — adversarial audit (as adjudicated)	**94 % — internal ceiling**
External residual: [H] values + feed representativeness	6	vendor gradation test, drop-weight A·b, sieve test, repeat belt-cuts, contractor design-curve reconciliation	→ 100 %

The residual six external points map one-to-one onto the chapter-10 triggers:

External residual	Closing trigger (chapter 10)	What moves when it closes
Smooth-roll n_comp / S_att [H]	T1 — vendor gradation test	C1 ratio, D6 margin, mode-F gap capability

A_j · b_j [H]	T2 — drop-weight test	CR.5011 product slope; the literature-tail scenario
Screen imperfection I [H]	T3 — product sieve test	every cut sharpness; KFS in-cut; convention nuance
Feed representativeness	T4 — repeat belt-cuts	the denominator of every yield; the quarry protocol
Design-curve divergence	T5 — contractor reconciliation	the KFS 80-vs-62 and FeedLime-split design questions

9.8 Honest limits

The 94 % is an internal ceiling by construction, and the report states plainly what no internal action can close. The [H] coefficient values themselves — the smooth-roll n_{comp} and S_{att}, the drop-weight $A_j \cdot b_j$, the screen imperfection — are hypotheses until the corresponding tests exist. The feed representativeness rests on a single belt-cut measurement (2026-08-08); the entire quantitative story pivots on it until repeats accumulate. And the model's original validation against the specification's chapter 9 is circular by construction (the reference curve was back-fitted to those figures): the verification program validates the model's mathematics and physics; real grounding comes only from feed measurements. These are precisely the chapter-10 triggers.

How to read the 94 % in one sentence each. What is *proven*: the code computes the documented formulas exactly (action 1), the formulas are the established ones correctly transcribed (actions 2 and 6), the model moves in the physically right directions under every probed condition (action 3), and the numerics are converged (action 5). What is *bounded*: the effect of every unmeasured coefficient, as an honest probability band with the project's central recommendation invariant across it (action 4). What is *pending*: the coefficient values themselves and the feed statistics — nothing internal can substitute for a test, and this report does not pretend otherwise.

10 Open items and external triggers

Everything that still needs a measurement, a vendor, a contractor, or a client arbitration — with what each one unlocks.

The design is internally verified to its ceiling; what remains is, by construction, external. Each trigger below names the measurement or counterpart that closes it and the exact model coefficient or decision it unlocks — when a result arrives, it lands in the data file first (the [H] coefficients refit with the archived calibration script), the test suite guards the physics, and the register gains a dated row. Nothing on this list blocks the purchase specifications of chapter 6 except where explicitly marked vendor-to-confirm.

10.1 Measurement and vendor triggers

#	Trigger	What it fixes / decides
T1	**Vendor gradation test** — smooth rolls on WANKOE 6/20 (trial trigger #1)	fixes n_{comp} = 1.8 [H] and S_{att} = 0.06 [H]; confirms the 1.5 mm mode-F gap capability and roll capacity at gap; confirms the 0.8-pt D6 grits margin under sliver regrind
T2	**Drop-weight test** (Q12)	fixes A_j = 60 / b_j = 0.8 [H]; decides whether the soft-limestone literature tail (b up to 3.0 → yield 21.8 %, landfill ~93 kt/y) is real
T3	**Screen-product sieve test** (Q3)	fixes I = 0.15 [H] and closes the disclosed convention nuance (spec-formula screens ≈ 2× sharper than a classic I = 0.15; classic imperfection 0.081)
T4	**Repeat belt-cuts** at the pivot	the single 2026-08-08 measurement is the pivot of every quantitative conclusion; repeats also drive the chapter-8 control protocol
T5	**Design-curve reconciliation with the engineering contractor** (RC7)	the PFD lists KFS at 80 t/h design where the engine computes 62.2 t/h wet at the adopted settings (51.35 at the pre-optimization settings, register 2026-08-12) — same family as the FeedLime/AgLime split inversion (61/39 vs the PFD's 42/58); one shared root: the design curve is finer than the measured feed

10.2 Design actions

#	Item	Status
A1	CR.5009 nip saturation — measured F_{80} 181 mm > 150 mm max nip (251 mm at the target curve)	standing alert; top-size control in the quarry conversation (ch. 8)
A2	BC.22 grits conveyor retrofit ~20 t/h — needed only by the 80 kt full-dryer extension case (16.7 > 15 t/h)	deferred with the extension provision (civil reserve kept)
A3	BE.40 fines conveyor	back within rating under C1; keep on the checklist until commissioning
A4	Final machine codification — SN.2x/CR.2x proposal for the C1 units to the engineering contractor	proposal pending
A5	Installed screen areas and motor sizes (all null data slots); linking-conveyor rating; sizing recommended for the extension per D1	awaiting design drawings

A6	Polishing bag filter for the sub-4 µm cyclone tail — absent from the flowsheet	design-review item (expert book, 2026-08-11)
A7	Documentation re-issue: zone-1.3 calculation document, DT-001 revision, web-UI zone-1.3 topology	pending

10.3 Open questions of the arbitration round

Q	Question	State
Q3	Imperfection-remap ratification + convention nuance	tied to the expert clarification note and the sieve test (T3)
Q4	KFS tolerance vs envelope (the nulled 15 % out-of-cut)	question pending
Q5	Spec §9.3 fines figure (19.9 t/h cannot close mass)	question pending
Q10	Repository relocation (deployment-safety caution)	open with the client
Q12	b_j for soft limestone	on hold pending the commissioned hardness note; T2 decides

10.4 Suggested sequencing

The natural order of closure: **T1 first** — the vendor gradation test gates the RC.2 purchase (the 1.5 mm capability and the D6 margin both hang on it) and it is the cheapest of the five; **T4 continuously** — repeat belt-cuts start delivering value with the second sample and drive the chapter-8 control protocol from day one; **T5 early** — the design-curve reconciliation with the engineering contractor is a meeting, not a test, and it retires the last systematic divergence between the two engineering records; **T2 and T3 opportunistically** — both refine coefficients whose uncertainty is already bounded by the chapter-9 Monte-Carlo band, so they sharpen the numbers without threatening the conclusions. The commissioned hardness note, when it arrives, is confronted with models M1–M8 and lands data-first like every other change.

Annex A Feed curves

The measured 2026-08-08 belt-cut (with H-FEED-1/2 completions) and the adopted quarry target, cumulative % passing.

Mesh (mm)	Measured (%)	Target (%)	Mesh (mm)	Measured (%)	Target (%)
0.063	0.07	0.05	15	41.30	34.58
0.1	0.12	0.08	**20**	**45.48**	**40.10**
0.125	0.17	0.11	25	47.58	43.56
0.25	0.42	0.29	31.5	49.75	46.42
0.5	1.06	0.73	35	50.74	47.41
1.0	2.67	1.84	40	52.00	48.66
1.5	4.53	2.90	50	56.19	50.76
1.7	5.31	3.47	63	60.52	53.86
2.0	6.53	4.22	80	65.00	58.34
2.8	9.86	6.39	100	71.05	62.53
4.0	14.87	9.89	125	76.40	67.47
5.0	18.82	13.02	160	78.82	73.81
6.3	23.50	16.63	200	81.00	77.52
8.0	28.71	21.15	250	90.02	79.71
10	33.57	25.84	320	100.00	100.00

Measured curve: belt-cut at the pivot 2026-08-08 (measured range 19–200 mm; fine tail completed by H-FEED-1, top size by H-FEED-2; moisture 7 % measured; model checks d50 32.3 vs measured 32 mm, d80 180.6 vs 180 mm). Target curve: shape-preserving rescale k = 1.4263 (chapter 8); log-linear interpolation between points in both cases. Control point 20 mm in bold.

Characteristic	Measured 2026-08-08	Quarry target
d50	32.3 mm	~46 mm
F80	180.6 mm	~251 mm (CR.5009 nip watch, §4.2)
Passing 20 mm (control point)	45.5 %	40.1 %
Passing 6.3 mm (zone-1.2 split driver)	23.5 %	16.6 %
Top size (H-FEED-2)	320 mm	320 mm (unchanged)
Moisture (wet basis)	7 %	7 % (assumed unchanged)

Annex B   Machine settings and purchase specifications

The adopted operating point and the rating of every wired machine, from data/default_parameters.json.

Machine	Type	Adopted settings	Rating / purchase spec
CR.5009	toothed roll crusher (z1.1)	g = 60 mm; n = 1.35; x80 = g	max nip 150 mm (saturated — alert); motor slot open
SR.5007	double-deck screen (z1.1)	35 / 20 mm; I = 0.15 [H]	required 6.80 / 7.15 m²; installed areas open
CR.5011	impact crusher HAZEMAG AP-S 1010 (z1.1)	v = 30 m/s; CSS = 30 mm	90 t/h real (wet basis, vendor); motor 132 kW
SR.5105	reclaim screen, single deck (z1.2)	a = 6 mm (FeedLime cut)	required 3.01 m²
SR.5111	1.7 mm open closing screen (z1.2)	a = 1.7 mm; I = 0.15 [H]	required 1.42 m²; loop conveyors 60 t/h (47 % used)
CR.5113	impact crusher, AgLime loop (z1.2)	v = 40 m/s; CSS = 1.0 mm	motor slot open (computed 87.4 kW installed)
SR.5115	1.7 mm closed-loop screen (z1.2)	a = 1.7 mm; I = 0.15 [H]	required 4.99 m²
DY.03	rotary dryer + burner BU.04 (z1.3)	m_out = 0.5 %	30 t/h AT OUTLET = 32.1 t/h wet feed (acquired, D2)
RC.1	smooth double roll, stage 1 (z1.3 C1)	g = 8 mm; λ 2.2, n 1.8, S_att 0.06 — all [H]	**32 t/h** (dry basis; client 2026-08-14)
RC.2	smooth double roll, stage 2 (z1.3 C1)	g = 3.4 mm (G) / 1.5 mm (F); same [H] set	**2 × 22 t/h** (dry basis); min gap 1.5 mm vendor-to-confirm
SC.A	double-deck recycle screen (z1.3 C1)	8 / 3.75 mm; I = 0.15 [H]	required 4.8 / 5.6 m² (worst mode)
SC.B	double-deck product screen (z1.3 C1)	2.0 / 1.5 mm; sliver regrind; DV.GF per mode	required 5.2 / 4.5 m² (worst mode)
SP.36	air classifier (UltraFin, optional)	cut 65 µm (Q6); η_cl 0.75 [H]	Φ at cut NOT CERTIFIED until measured
CL.38	cyclones (UltraFin)	v_in = 15 m/s; b = 0.14 m (Stairmand, to confirm)	d50 = 4.2 µm; bag filter missing (design review)

As-built zone-1.3 units SN.21 and ML.26 remain in the data as the selectable "as-built" variant for history and epoch tests. ML.26 coefficients (λ 4.11, S_att 0.171, n_comp 0.86) are the fitted H-M7 hypotheses. Calibration constants: Wi 12.54 kWh/t [ref.], η_m 0.75, A_j 60 / b_j 0.8 [H], f0 0.347 [H], f_p 1.2 [H], I_dry 0.15 [H], I_rain 0.9 [H], dryer η_th 0.6 / I_ev 45 [H], λ_air 0.5 [H], N_e 5 [H], ρ_p 2 700 kg/m³ [H].

B.1   Stockpile capacities (PFD, data only)

Yard	Capacity (t)	Role at the adopted plan

KFS Fines 0/20 (SP.5014)	10 000	inter-zone buffer; nets to zero over the year (landfill absorbs the excess)
KFS 20/35 (SP.5015)	5 000	KFS shipping buffer (weekly kiln rhythm)
Feed 0-6/20 (SP.5109)	6 000	FeedLime weekly buffer — seasonal-swing watch (§5.5)
AG 0/1.7 (SP.5118)	28 000	AgLime campaign buffer

B.2 Engine numerical settings

Key	Value	Meaning
computation_grid_refinement	2	×2 converged grid (57 meshes); 1 reproduces spec-era numerics
loop_relative_tolerance	10^{-6}	fixed-point convergence (10^{-9} verified identical, action 5)
loop_max_iterations / max_circulating_ratio	200 / 10	divergence self-declared, never silently reported
balance_relative_tolerance	10^{-3}	assertion threshold; realized gaps < 2×10^{-6}
extension_meshes_mm	250, 320	top-size completion meshes (H-FEED-2)

Annex C Replayable scripts

The verification chain is replayable without any assistant — one command each, from the repository root.

Script	What it replays
`scripts/verify_zone11_independent.py`	Action 1 — the blind stdlib re-derivation of the whole zone-1.1 chain from documentation only; prints the 25-scalar + 29-point concordance table. Run: `python scripts/verify_zone11_independent.py`
`scripts/monte_carlo_kfs_yield.py`	Action 4 — the 200-draw seeded Monte-Carlo band over the five [H] ranges (seed 7); prints P10/P50/P90 KFS Yield and implied landfill. Run: `PYTHONPATH=src python scripts/monte_carlo_kfs_yield.py`
`scripts/numerical_robustness.py`	Action 5 — grid ×2/×4 refinement and loop-tolerance study; prints the convergence table behind the ×2 grid adoption. Run: `PYTHONPATH=src python scripts/numerical_robustness.py`
`scripts/stress_test.py`	Action 3 — the 580-run stress campaign (500 seeded random draws + settings extremes + degenerate feeds + brutal rates); asserts balances, monotonicity, finiteness. Run: `PYTHONPATH=src python scripts/stress_test.py`

The full automated suite (138 tests, including the literature anchors, the physical invariants, the chapter-9 epoch pins and the C1/zone-1.3 baselines) runs with `python -m pytest tests/ -q`.

C.1 Reproducing this report's engine runs

Every table in chapters 4–7 replays from the repository root with the public API (`PYTHONPATH=src`, or the package installed):

Run	Call
Mode-G photo (chapters 4, 5, 6)	`run_scenario(load_parameters())`
Mode-F photo (chapter 6)	`run_scenario(load_parameters(overrides={"default_scenario": {"zone_1_3_mode": "F"}}))`
As-built counterfactual (§6.1)	`… overrides={"default_scenario": {"zone_1_3_variant": "as-built"}}`
Mode-F gap counterfactual (§6.7)	`… overrides={"default_scenario": {"zone_1_3_mode": "F"}, "machines": {"RC.2": {"mode_F_gap_mm": 2.8}}}`

Rain photo (§5.4)	… overrides={"default_scenario": {"weather": "rain"}}
2C campaign photo (§5.4)	… overrides={"default_scenario": {"zone_1_2_mode": "2C"}}
Annual plan, measured feed (chapter 7)	run_required_hours(load_parameters())
Annual plan at the quarry target (chapters 7, 8)	run_required_hours(load_parameters(overrides={"feed_product": {"cumulative_passing_curve": <target curve JSON>}, "production_targets": {"AgLime 0/1.7": {"target_t_per_year": 108000, "market_cap_t_per_year": 108000}}}))

Provenance — every figure in this document is an engine execution or a dated decision-register quotation: run_scenario (mode-G and mode-F photos, as-built counterfactual, mode-F 2.8 mm gap counterfactual) and run_required_hours (two-mode plan, 2C campaigns) at the converged ×2 grid, defaults plus the quarry-target overrides (quarry-target-curve-20pct-margin.json, AgLime baseline 108 kt); repository branch wankoe-python-model, 2026-08-14. Hypothesis coefficients flagged [H] throughout; UltraFin figures NOT CERTIFIED until Φ is measured. Verification dossier: docs/design/zone11-verification (actions 1–6, meta-score 94 %). Decision register: docs/spec-conformity-matrix.md. Quarry curve by bisection (scripts archived, replayable without the assistant). — **by NOEZYS**.